Agentic AI Roadmap Stages: Beginner → Intermediate → Advanced

Here's a suggested stage-by-stage learning path.

Stage	Time	Focus Areas / Skills	Free Resources
Stage 1: Fundamentals	~2-4 weeks	Python, Math foundations, ML basics, Deep Learning intro	- Free AI & ML Roadmap from Scratch (GitHub: aadi1011) — includes free courses/modules. GitHub - MIT OpenCourseWare linear algebra lectures / Khan Academy / 3Blue1Brown "Essence of linear algebra" - Andrew Ng's Machine Learning course (Coursera) — has audit option free - DeepLearning.AI's "Neural Networks and Deep Learning" (Coursera) — audit free options
Stage 2: Generative AI, LLMs & Language Models	~2-3 weeks	Transformers, LLMs fundamentals, generative models (text, images), prompt engineering	- Hugging Face's free course on Transformers - Free lectures / tutorials on GANs, VAEs - "Generative AI – A Way of Life" from Analytics Vidhya (free) ⟶ part of their Agentic AI path. Analytics Vidhya+1 - YouTube tutorials (e.g. "Transformers explained")
Stage 3: Agentic Concepts	~2-3 weeks	What makes an agent: state, memory, planning, tool use, decision loops, ReAct / Tree-of-Thoughts / reflection patterns, multi-agent systems	- "How to Become an Agentic AI Expert in 2025" (Analytics Vidhya) — they lay out a weekly learning path. Analytics Vidhya - The "Agentic AI: A Self-Study Roadmap" on KDnuggets. KDnuggets - GitLab's agentic AI guides & resources. about.gitlab.com
Stage 4: Tooling & Frameworks	~2-3 weeks	LangChain, LangGraph, AutoGen, LlamaIndex (or similar), RAG (Retrieval-	- Tutorials & docs from LangChain site (free) - Free blog / YouTube

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		Augmented Generation), integrating external APIs, memory modules	guides: building agentic AI with LangChain etc. - Use LlamaIndex open-source docs / examples - "Free AI & ML Roadmap from Scratch" includes Generative AI / RAG modules. GitHub
Stage 5: Building Projects	~3-4 weeks	Build small agents, then more complex ones: e.g. personal assistant agent, multi-step planner, multi-agent collaboration, maybe environment interaction	- Use OpenAI free tier / free LLMs (or open-source ones) - Build agents with LangChain + Python - Projects from the Analytics Vidhya path. Analytics Vidhya - GitHub repositories / community projects (e.g. from "AI & ML Roadmap from Scratch") to contribute or replicate. GitHub
Stage 6: Safety, Ethics, Alignment & Advanced Topics	~2 weeks (ongoing)	Ensuring agent behaviour is aligned, learning from experience (lifelong learning), robustness, evaluation metrics, adversarial considerations, governance	- Paper: *Lifelong Learning of Large Language Model based Agents: A Roadmap* (survey) arXiv - Read about alignment, constitutional AI (free resources like blog posts, open papers) - Explore recent works like "LightAgent" (an agentic framework) which is open source. arXiv - Keep up with recent papers via arXiv / ML conferences
Stage 7: Deployment, Scaling, Multi-Agent & Environment Interaction	~3-4 weeks	Deploying agents (cloud, containers), integrating sensors or external systems, multi-agent systems, collaboration, persistence (memory over long time), RL or environment based agents, evaluation & monitoring	- Use free tiers of AWS / GCP / Azure to deploy simple agents - Explore open-source frameworks like LightAgent. arXiv - Build multi-agent projects (e.g. agents that communicate or coordinate) - Use "AWorld" (recent system for agent-

Stage	Time	Focus Areas / Skills	Free Resources
			environment interaction) as case study. arXiv

Hope you found this guide helpful! If you want a 1:1 guidance from me, you can book a slot here.

Feel free to send your feedback on Instagram.

All the best!